RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Additional Practice Questions for Chapter S1B: Probability

1 For events A and B, it is given that $\mathrm{P}(A|B') = \frac{2}{3}$, $\mathrm{P}(A \cap B) = \frac{1}{10}$ and $\mathrm{P}(B) = \frac{2}{5}$. Find

(i) $\mathrm{P}(A \cap B')$, [1]

(ii) $\mathrm{P}(A)$. [1]

(iii) State whether A and B are independent, giving a reason for your answer. [1]

[**(i)** $\frac{2}{5}$ **(ii)** $\frac{1}{2}$ **(iii)** A and B are not independent]

2 For events A and B, it is given that $\mathrm{P}(A) = 0.7$, $\mathrm{P}(B|A') = 0.8$ and $\mathrm{P}(A|B') = 0.88$. Find

(i) $\mathrm{P}(B \cap A')$, [1]

(ii) $\mathrm{P}(A' \cap B')$, [2]

(ii) $\mathrm{P}(A \cap B)$. [3]

[**(i)** 0.24 **(ii)** 0.06 **(iii)** 0.26]

3 For events A and B, it is given that $\mathrm{P}(A') = 0.3$, $\mathrm{P}(B') = 0.6$ and $\mathrm{P}(B \mid A') = 0.4$.

(i) Find $\mathrm{P}(A' \cap B)$. [2]

(ii) Find $\mathrm{P}(A \cap B)$. [2]

(iii) Determine whether A and B are independent events. [1]

For a third event C, it is given that A and C are mutually exclusive, and $\mathrm{P}(B \cap C) = 0.1$.

(iv) Find the possible range of values of $\mathrm{P}(C)$. [2]

[**(i)** 0.12 **(ii)** 0.28 **(iii)** independent **(iv)** $0.1 \le \mathrm{P}(C) \le 0.28$]

4 In each turn of a game, a player throws a fair die repeatedly until he obtains a number that has already appeared in that turn; the player must therefore have at least 2 throws but not more than 7 throws.

(i) Find the probability that the player has fewer than 4 throws in a turn.

(ii) Find the conditional probability that the player has exactly 2 throws in a turn, given that he has fewer than 4 throws in that turn.

(iii) Find the probability that the player has exactly 5 throws in a turn.

[**(i)** $\frac{4}{9}$ **(ii)** $\frac{3}{8}$ **(iii)** $\frac{5}{27}$]

5 A company buys p % of its electronic components from supplier A and the remaining $(100 - p)$ % from supplier B. The probability that a randomly chosen component supplied by A is faulty is 0.05. The probability that a randomly chosen component supplied by B is faulty is 0.03.

(i) Given that $p = 25$, find the probability that a randomly chosen component is faulty. [2]

(ii) For a general value of p, the probability that a randomly chosen component that is faulty was supplied by A is denoted by $\mathrm{f}(p)$. Show that $\mathrm{f}(p) = \dfrac{0.05p}{0.02p + 3}$.

Prove by differentiation that f is an increasing function for $0 \le p \le 100$, and explain what this statement means in the context of the question. [6]

[**(i)** 0.035]

6 A teacher conducted a survey on a large number of students to determine the choice of colours for painting the school hall from 3 colour options of white, green and blue. Of the students surveyed, 40% were boys and 60% were girls. Of the boys, 50% chose white, 20% chose green and the rest chose blue. Of the girls, 25% chose white, 45% chose green and the rest chose blue.

Draw a probability tree diagram to illustrate the above information. [1]

(i) One student is randomly selected. Find the probability that the student chose white. [1]

(ii) Two students are randomly selected. Find the probability that the two students are of the same gender or chose different colours (or both). [3]

(iii) Three girls are randomly selected. Find the probability that exactly 1 girl chose white, given that none of them chose blue. [3]

[**(i)** 0.35 **(ii)** $\dfrac{1067}{1250}$ **(iii)** $\dfrac{1215}{2744}$]

7 The probability that a hockey team wins any match is $\dfrac{1}{2}$ and the probability that it loses any match is $\dfrac{1}{6}$. Three points are awarded for a win, one point for a draw and no point for a defeat. The team plays four matches and the outcome of each match is independent of the other matches.

(i) Show that the probability that the team has exactly one draw and exactly one defeat is $\dfrac{1}{6}$. [1]

Find the probability that the team

(ii) wins the first match and goes on to win exactly one other match, [3]

(iii) wins exactly one match, given that it obtains four points. [4]

[**(ii)** $\dfrac{3}{16}$ **(iii)** $\dfrac{9}{11}$]

8 3 machines A, B and C produce 25%, 35% and 40% respectively of the golf balls manufactured by a factory. These balls are either yellow or white. Of the balls produced by A and B, 20% and 30% respectively are yellow. It is known that the probability of picking a yellow ball is 0.355.

(a) If 3 balls are picked randomly, find the probability that

(i) at least 2 are produced by machine B, [2]

(ii) at least 1 is a yellow ball, [2]

(b) at least 1 is yellow given that all the balls picked are from machine A. [2]

(b) Due to malfunction of machine A, 10% of the balls produced by machine A is faulty and the event of a ball produced by A is faulty is independent of its colour. Find the proportion of white balls that are faulty and produced by machine A. [3]

[**(a)(i)** 0.28175 **(ii)** 0.732 **(iii)** 0.488 **(b)** 0.0310]

9 **(a)** A student threw a fair die four times and recorded the number obtained for each throw. Find the probability that

(i) at least two of the four numbers obtained are the same, [2]

(ii) exactly two of the four numbers obtained are the same given that the number 6 appeared at least once. [4]

(b) Events A and B are such that $\mathrm{P}(B') = \frac{1}{2}$, $\mathrm{P}(A' \cap B') = \frac{7}{20}$ and $\mathrm{P}(B|A) = \frac{4}{7}$. Find $\mathrm{P}(A)$. [3]

[**(a) (i)** $\frac{13}{18}$ **(ii)** $\frac{360}{671}$ **(b)** $\frac{7}{20}$]

10 A teacher brings 4 black, 3 blue, 2 red and 1 green markers to the classroom for each of his lessons. Unknown to him, the probabilities that a black, blue, red or green marker is out of ink are $p, \frac{1}{2}p$, $\frac{1}{4}p$ and $\frac{1}{8}p$ respectively, where $0 < p < 1$.

(i) Find, in terms of p, the probability that a randomly chosen marker from his set of ten markers is out of ink. [2]

(ii) In the classroom, the first marker he tries out is out of ink. Find, in terms of p, the probability that the next marker he tries out is a red marker that is also out of ink. [3]

(iii) After numerous lessons, the teacher realized that, in general, the first marker he tries out works at least 7 out of 10 times. Find the range of possible values of p. [2]

[**(i)** $\frac{49}{80}p$ **(ii)** $\frac{47}{882}p$ **(iii)** $0 < p \leq \frac{24}{49}$]

11 A public opinion poll surveyed a sample of 1000 voters. The table below shows the number of males and females supporting Party A, Party B and Party C.

	Party A	**Party B**	**Party C**
Male	200	130	70
Female	250	300	50

(a) One of the voters is chosen at random. Events A, C and M are defined as follows:

A : The voter chosen supports Party A.
C : The voter chosen supports Party C.
M : The voter chosen is a male.

Find
(i) $\mathrm{P}(A \mid M)$,
(ii) $\mathrm{P}(M' \cap C')$.
Determine whether A and M are independent. [4]

(b) It is given that in the sample, 20% of Party A supporters, 30% of Party B supporters and 5% of Party C supporters are immigrants.

(i) One of the voters selected from the sample at random is an immigrant. What is the probability that this voter supports Party A? [2]

(ii) Three voters are chosen from the sample at random.
Find the probability that there is exactly one immigrant voter who supports Party C or exactly one female who supports Party A (or both). [4]

[(a) (i) $\frac{1}{2}$ **(ii)** $\frac{11}{20}$ **(b)(i)** 0.4 **(ii)** 0.434]

12 [In this question, give all your answers correct to 3 decimal places.]
A delegation of 20 athletes comprises 9 basketball players, 6 volleyball players and 5 tennis players. Among them are 5 male basketball players, 6 male volleyball players and 2 male tennis players.
A committee of four people is to be formed from these 20 athletes.

(i) Find the probability that exactly three basketball players and one tennis player are chosen. [2]

(ii) Find the probability that 2 female athletes are chosen given that at least one volleyball player is chosen. [4]

(iii) Determine if the 2 events "2 female athletes are chosen" and "at least one volleyball player is chosen" are independent, justifying your answer. [2]

[(i) 0.087 **(ii)** 0.311]

13 A machine-operated car wash offers four different options as follows:

Option	Description	Duration
A	Wash only	2 minutes
B	Wash and dry	3 minutes
C	Bubble-wash and dry	3 minutes
D	Steam-wash and dry	4 minutes

A driver makes his choice and then proceeds to the car wash. The probability of a driver choosing Option A, Option B, Option C or Option D is given to be $\frac{5}{12}$, a, ar and ar^2 respectively, where a and r are positive constants and $r \neq 1$.

At a particular instant, there are three cars queuing to use the car wash.

(i) Show that the probability that at least one of the three drivers has chosen option C is given by $1-(1-ar)^3$. [1]

(ii) By considering the sum of the probabilities for all the options, deduce that $0 < a < \frac{7}{12}$. [3]

For the rest of the question, take $a = \frac{1}{3}$ and $r = \frac{1}{2}$.

A fourth car, car X joins the queue at the instant when the first car enters the car wash. Assuming that no time is lost between the time a car leaving the car wash and the next car starting the wash, find the probability that

(iii) it will be at least 10 minutes before car X enters the car wash and at least one of the previous three drivers chose Option C, [2]

(iv) it will be at least 10 minutes before car X enters the car wash, given that at least one of the previous three drivers chose Option C. [2]

State a necessary assumption for your calculations in **(iii)** and **(iv)**. [1]

[**(iii)** $\frac{11}{288}$ **(iv)** $\frac{33}{364}$]

14 **(a)** Events A and B are such that $\mathrm{P}(A)=\frac{1}{3}$, $\mathrm{P}(B \mid A)=\frac{1}{3}$ and $\mathrm{P}(A' \cap B')=\frac{1}{6}$. Find

(i) $\mathrm{P}(A \cup B)$, [2]

(ii) $\mathrm{P}(B)$. [3]

(b) A man writes 5 letters, one each to A, B, C, D and E. Each letter is placed in a separate envelope and sealed. He then addresses the envelopes, at random, one each to A, B, C, D and E.

(i) Find the probability that the letter to A is in the correct envelope and the letter to B is in an incorrect envelope. [1]

(ii) Find the probability that the letter to A is in the correct envelope, given that the letter to B is in an incorrect envelope. [3]

(iii) Find the probability that both of the letters to A and B are in incorrect envelopes. [5]

[**(a)(i)** $\frac{5}{6}$ **(ii)** $\frac{11}{18}$; **(b)(i)** $\frac{3}{20}$ **(ii)** $\frac{3}{16}$ **(iii)** $\frac{13}{20}$]

15 Three cards are drawn from an ordinary pack of 52 cards, at random and without replacement. Cards drawn have the following values:

Aces score 1 point,
Tens, Jacks, Queens and Kings score 10 points, and
Cards from two to nine score as many points as the numbers they carry (i.e. twos score 2 points, threes score 3 points, and so on).

Find the probabilities that

(i) exactly two cards each scoring more than 5 points are drawn, [2]

(ii) all three cards are of different suits, [3]

(iii) total score of the three cards is more than 28 points given that all three cards are of different suits. [4]

[**(i)** $\frac{496}{1105}=0.449$ **(ii)** $\frac{169}{425}=0.398$ **(iii)** $\frac{112}{2197}=0.0510$]

16 For events A and B, it is given that $\mathrm{P}(A)=0.6$, $\mathrm{P}(A\cup B)=0.7$ and $\mathrm{P}(B\mid A)=0.3$.

(i) Show that $\mathrm{P}(B)=0.28$ and determine with a reason, if A and B are independent. [3]

(ii) An event C is such that A and C are mutually exclusive and $\mathrm{P}(C)>0.25$. Find the range of values of $\mathrm{P}(B'\cap C)$. [3]

[**(i)** not independent **(ii)** $0.15<\mathrm{P}(B'\cap C)\leq 0.3$]

17 A room contains n randomly chosen people.

(a) Assume that a randomly chosen person is equally likely to have been born on any day of the week. The probability that the people in the room were all born on different days of the week is denoted by P.

(i) Find P in the case $n=3$. [2]

(ii) Show that $P=\dfrac{120}{343}$ in the case $n=4$. [1]

(b) Assume now that a randomly chosen person is equally likely to have been born in any month of the year. Find the smallest value of n such that the probability that the people in the room were all born in different months of the year is less than $\dfrac{1}{2}$. [4]

(c) Assume now that a randomly chosen person is equally likely to have been born on any of the 365 days in the year. It is given that, for the case $n=21$, the probability that the people in the room were all born on different days of the year is 0.55631, correct to 5 places of decimals. Find the smallest value of n such that the probability that at least two of the people were born on the same day of the year exceeds $\dfrac{1}{2}$. [5]

[**(a)(i)** 30/49 **(b)** 5 **(c)** 23]

THE END